

Cost-Sensitive Knowledge Discovery for 30-Day Hospital Readmission in Diabetic Patients

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Abstract

Hospitals unable to effectively predict which patients are likely to require short-term readmission are missing signs of severity, may be misallocating resources and, in some jurisdictions, facing financial penalties. However, while modern predictive models frequently rely on opaque machine learning methods to maximize raw accuracy, clinical settings demand transparent and actionable insights. This project addresses this need by applying Knowledge Discovery techniques to the Diabetes 130-US Hospitals dataset. Emphasizing a cost-sensitive evaluation framework, we optimize algorithms to minimize the financial burden of missed readmissions while prioritizing interpretable models like Decision Trees and Logistic Regression. Our findings reveal that balancing methods (SMOTE) and appropriate evaluation metrics (AUC-ROC) are essential given the 11% minority class imbalance, while SHAP values and unsupervised analysis can lead to a balance between high-performing black-box models and clinically-useful interpretable models. Ultimately, we propose transitioning from simple probability scores to clinical risk triggers, enabling targeted, cost-effective preventative care, and provide analysis and recommendations for hospital decisions. All code and analysis notebooks are made publicly available at <https://github.com/chrisindris/CP8305FP/>.

I. INTRODUCTION

THE shift toward value-based healthcare and the Affordable Care Act’s Hospital Readmissions Reduction Program (HRRP) [1] heavily penalize hospitals for excessive 30-day patient readmissions. This paradigm shift makes early risk detection a primary operational challenge for medical facilities. Although modern predictive modeling often relies on opaque machine learning techniques (such as XGBoost) to maximize raw accuracy [2], clinical environments require transparent, actionable insights. Hospital administrators may be wary of medical decisions not made by medical professionals, and physicians may be hesitant to act on black-box risk scores for malpractice avoidance. This project selects the Diabetes 130-US Hospitals dataset [3] as a well-known dataset including readmission decisions which includes a useful number of attributes for medications and diagnoses.

By applying interpretable Knowledge Discovery (KD) algorithms and cost-sensitive evaluation metrics, our objective is to identify the core clinical drivers of early readmission and provide decision-makers with scalable, human-readable intervention strategies. Specifically, we focus on identifying high-risk diabetic patients at the time of discharge, enabling targeted and cost-effective preventive care. Our approach explicitly tackles the methodological gap identified by Hu & Sokolova [4] regarding the necessity of aggressively addressing minority class imbalance, while adopting the business context proposed by Graham et al. [5], evaluating model success through the lens of HRRP financial penalties rather than mere statistical accuracy. Previous works may not have focused on context-appropriate punishment for false negatives or inpatient vs. outpatient medications [6], or lack focus on transparent models and the production of rule sets for clinical use [7]. It is our hope that through supervised and unsupervised analysis, these relationships can be studied for actionable insights and to spur on further research into the complex drivers of readmission risk in diabetic patients.

II. DATA

The foundation of our analysis is the Diabetes 130-US Hospitals dataset (1999–2008), which covers 10 years of clinical care from the Cerner Health Facts database. The target is defined as a binary classification problem: grouping “NO” (no readmission) and “>30 days” readmission into the negative (low-risk) class, while isolating < 30 days as the positive (high-risk) class. This framing aligns our model directly with the 30-day HRRP financial penalty window, and previous research [1].

The dataset introduces several significant hurdles, notably an extreme class imbalance, where the positive class represents only 11% of the records. It is also plagued by sparsity and high cardinality, and large size after one-hot encoding; some attributes have hundreds of classes. Our preprocessing pipeline thus included the following.

- 1) **Data Cleaning:** Missing values encoded as “?” were replaced with NaN, though in some attributes such as medical tests the “missing” value is indicative of a medical test not being ordered, and is thus informative. Variables like `weight` were dropped entirely due to ~97% sparsity. Features like `payer_code` and `medical_specialty` were dropped after initial review following literature precedents.
- 2) **Removing Leakage:** Records with terminal or hospice discharge outcomes—where 30-day readmission is clinically impossible—were removed to prevent bias. Furthermore, only the first encounter per patient was retained to preserve independence among observations. Identifier columns (`encounter_id` and `patient_nbr`) were dropped prior to modeling.

3) **Feature Engineering:** High cardinality ICD-9 diagnosis codes were collapsed into nine high-level categories (e.g., Diabetes, Circulatory, Respiratory) as suggested by Strack et al. [1]. Furthermore, we engineered novel variables capturing medication dynamics and healthcare utilization: `total_drug_changes` (count of medication prescription adjustments), `comorbidity_count`, and prior visit summaries.

The numerical attributes independently have weak correlations with the target variable (Fig. 1). To delve deeper into the interplay of clinical prescriptions, we leveraged Association Rules. In medical datasets facing extreme sparsity and an overwhelming number of concurrent variables, mapping association rules provides immense value. These rules isolate concurrent prescription regimens (e.g., Insulin paired with Metformin) and comorbidities that frequent high-risk patient subpopulations. More importantly, association rule mining distills complex interactions into transparent logic (rules structured as IF conditions THEN outcome), delivering immediately human-readable intelligence that clinicians can act upon during discharge to preempt readmission.

As an initial exploratory step, we extracted the top 5 global association rules (Table I) sorted by lift to identify dominant patient subgroups. Notably, the strongest dataset-wide rules universally featured `readmitted=0` combined with multiple “No” medication prescriptions as antecedents, leading to the absence of other medications as consequents. Rather than explicitly isolating high-risk readmission patterns, these highest-confidence global rules effectively profile stable, low-acuity patients (e.g., normal/unmeasured A1C and glucose, with absent diabetic medication regimens). The inability of global association rules to explicitly capture the high-risk minority class emphasizes the complexity of the readmission condition, motivating our subsequent supervised and clustering-based approaches to uncover the true high-risk boundaries.

TABLE I: Top 5 Global Association Rules by Lift

Antecedents (IF)	Consequents (THEN)	Support	Conf.	Lift
readmitted=0, diabMed=No, A1C=None, max_glu=None, glipizide=No, glyburide=No	metformin=No, pioglitazone=No, rosiglitazone=No	0.171	1.000	1.407
readmitted=0, diabMed=No, gender=F, A1C=None, max_glu=None, insulin=No...	glipizide=No, pioglitazone=No	0.095	1.000	1.225
readmitted=0, diabMed=No, gender=M, race=Cauc, change=No, max_glu=None...	glipizide=No, pioglitazone=No	0.067	1.000	1.225
readmitted=0, diabMed=No, gender=M, race=Cauc	glipizide=No, pioglitazone=No	0.072	1.000	1.225
readmitted=0, diabMed=No, gender=F, change=No, A1C=None, insulin=No...	glipizide=No, pioglitazone=No	0.102	1.000	1.225

Note: Characteristics strongly associated with non-readmission mostly highlight low-acuity profiles.

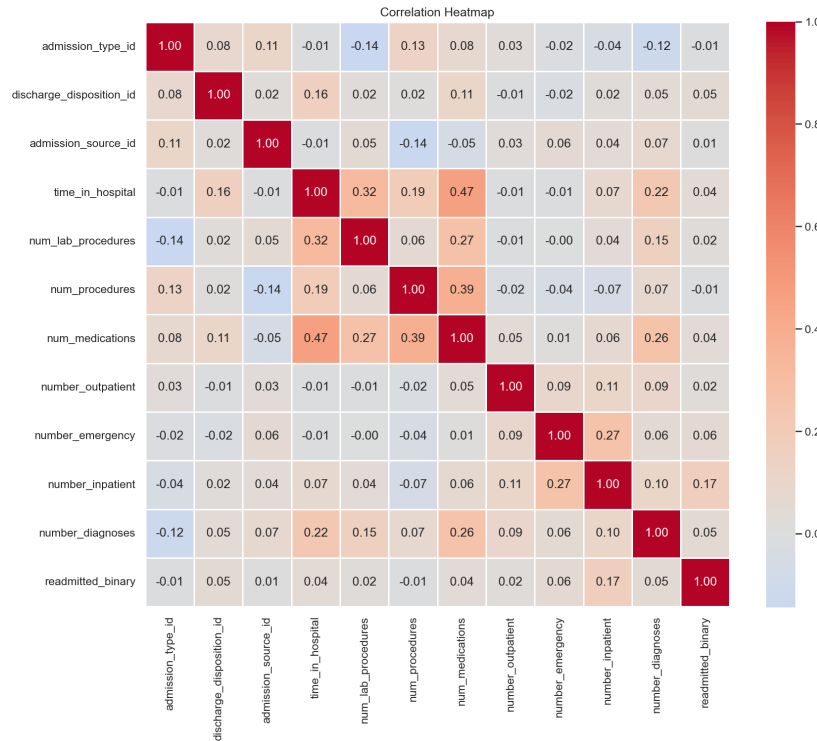


Fig. 1: Correlation heatmap revealing the relationships between key engineered clinical features and the readmission target variable. While this can indicate where to seek future research for causal links, we note that there is limited correlation with the target variable.

III. MODEL

To bridge the gap between predictive accuracy and actionable Knowledge Discovery, our modeling framework benchmarks interpretable algorithms (Logistic Regression, Decision Trees) against strong opaque performers (LightGBM [8], XGBoost [9], Random Forest, Gradient Boosting).

To fairly assess performance across algorithms, we employed a 10-Fold Stratified Cross-Validation nested scheme. The outer loop preserves genuine model evaluation, while an inner loop (5-Fold Stratified CV) facilitates hyperparameter tuning (GridSearchCV). Over-sampling the minority class is achieved locally within the CV partitions using the Synthetic Minority Over-sampling Technique (SMOTE) [10], protecting the validation sets from data leakage. Standardized continuous attributes are likewise scaled exclusively on train folds.

Because accuracy is heavily impacted by the dataset imbalance, evaluating the predictive quality based on a conventional accuracy metric is highly deceptive. Instead, models were assessed via AUC-ROC and AUPRC. Furthermore, we explicitly introduced a Cost-Sensitive Evaluation framework derived from business logic:

- **False Negatives:** Missed readmissions carry a severe penalty (e.g., -\$15,000) under the HRRP framework.
- **False Positives:** Unnecessary interventions carry a mild operational cost (e.g., -\$200).
- **True Positives:** Net a preventative benefit (-\$2,000 baseline cost for intervention instead of full readmission).

Using these assigned magnitudes, we optimize the decision thresholds. Rather than defaulting to a 0.5 probability cutoff, adjusting the threshold explicitly minimizes expected total costs.

IV. ANALYSIS AND FINDINGS

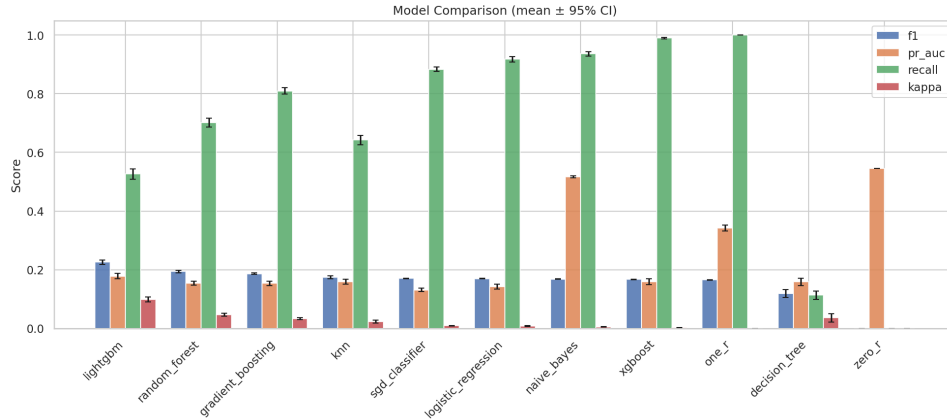


Fig. 2: Performance of different models on various metrics, after applying techniques to address class imbalance. LightGBM achieved the best overall performance.

TABLE II: Cross-Validation Performance Results (Mean Metrics). *Note:* Default threshold metrics for models. AUC-ROC remains robust despite poor F1 calibration.

Model	F1-Score	AUC-ROC	PR AUC	Balanced Accuracy
SGD Classifier	0.208	0.620	0.146	0.588
Gradient Boosting	0.069	0.640	0.145	0.511
KNN	0.062	0.530	0.111	0.508
LightGBM	0.047	0.622	0.155	0.510
Random Forest	0.000	0.618	0.159	0.500

Our investigation confirmed that the intense dataset imbalance (approximately an 11:89 ratio) makes vanilla model performance misleading; zero-rule or baseline models deceptively yield 89% accuracy without predicting a single hospitalization. The initial modeling phases underscored that the dataset is fundamentally challenging and not linearly separable. The complex, overlapping clinical profiles of the patient population meant that linear instances struggled to define clear boundaries separating the minority class. The task imbalance and lack of separability makes this a challenging benchmark, though the imbalance-aware AUC-ROC remained a useful metric for which LightGBM performed best (Fig. 2, Fig. 3 and Table II).

To combat this, we deeply invested in strategies to forcefully expose the models to the high-risk patient subset. Our pipeline explicitly tested several balancing strategies, eventually standardizing on a combination of built-in algorithm class weighting alongside fold-local over-sampling (via SMOTE). This twofold approach empowered the algorithms to learn the rare profiles safely inside the nested validation loops without causing data leakage. Furthermore, the necessary one-hot encoding of clinical categories temporarily ballooned the feature space to over 2,400 columns. To restrict model complexity and

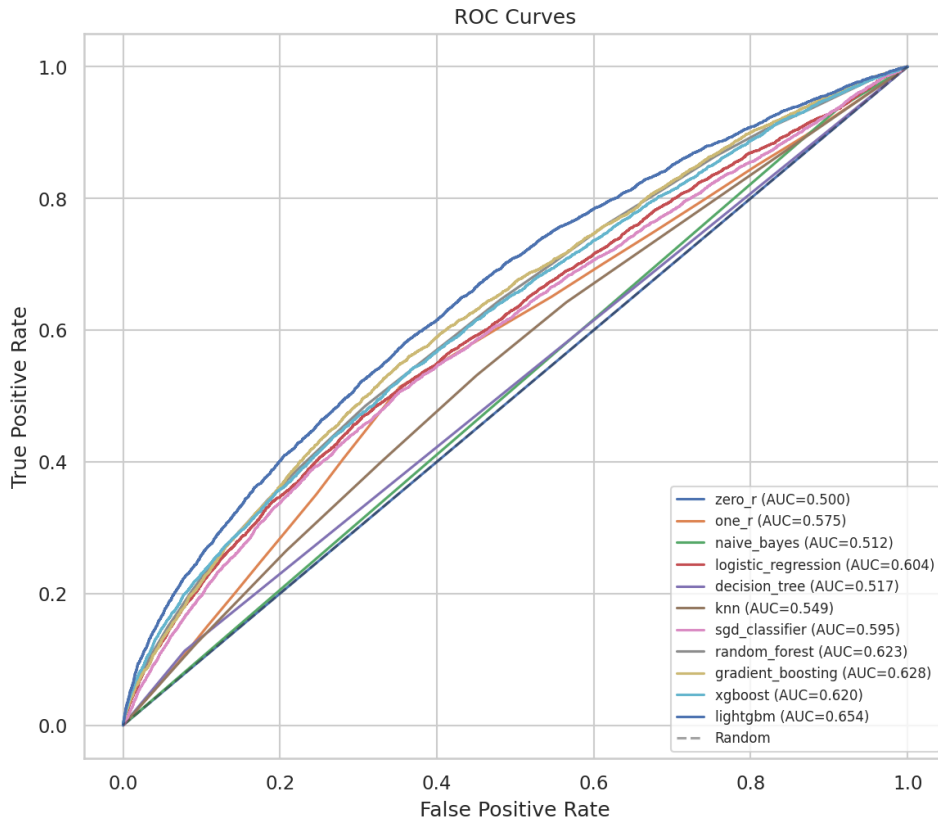


Fig. 3: Receiver Operating Characteristic (ROC) curves highlighting the performance of the various tuned models.

computational overhead without surrendering fidelity, we heavily utilized fold-local dimensionality reduction. We implemented L1-regularized selection and an ensemble-backed `SelectFromModel` approach, alongside PCA [11] and `SelectKBest`. This systematically isolated the most predictive factors while aggressive algorithmic pruning shed the uninformative, sparse clinical noise.

However, moving past generic metrics provided more actionable insights. The AUC-ROC proved robust in handling class imbalance. Notably, LightGBM achieved the highest cross-validation F1-score (0.226) and AUC-ROC (0.654), statistically outperforming baselines. Additionally, the LightGBM model exhibited strong probability calibration for the minority class, achieving a Brier Score of 0.0789 on the holdout test set. After proper tuning and feature elimination, models began identifying valid correlative relationships in patient covariates.

To ensure clinical interpretability of the black-box algorithms, we leveraged explainability tools including SHAP [12] summary values (Figure 4), permutation importance (Figure 5), and feature importance (Figure 6). We found that features detailing the count of concurrent medications (`num_medications`), changes in specific diabetic prescriptions, and historical healthcare utilization (inpatient and emergency visits) heavily govern short-term risk.

The final model evaluation on a hold-out test set perfectly illustrated the deceptive nature of this highly imbalanced dataset. The model achieved a seemingly strong raw accuracy of 90.96%. However, utilizing the default decision threshold, the recall for the minority class (actual readmissions) plummeted to 0.02, leading to an F1-score of merely 0.04 for the positive class (Figure 7). Though the model's precision was high (0.92), capturing only 2% of actual readmissions renders the model clinically ineffective under a standard probability threshold.

Critically, when implementing the cost-sensitive framework, the default 0.5 probability threshold severely penalized the model leading to missed readmissions. The cost-optimal evaluation highlighted extreme penalty asymmetries, and tuning the decision threshold strictly below 0.5 (specifically optimized to 0.10 for recall, as shown in Figure 8) radically improved the capture of minority class instances. Taking a larger volume of False Positives (spending minor sums on preventative care or phone calls) vastly outweighs missing a True Positive (a penalized HRRP readmission).

Based on these findings, hospital administration should integrate these interpretable risk triggers directly into their Electronic Health Record (EHR) discharge workflows. Rather than presenting clinicians with an abstract probability score, the system should actively flag patients who exhibit high-risk characteristics—such as complex medication dynamics or high historical utilization—using the optimized cost-sensitive threshold. Once flagged, standard low-cost preventative protocols, such as scheduled outpatient follow-up calls or comprehensive medication reconciliations by a pharmacist, can be automatically triggered. This

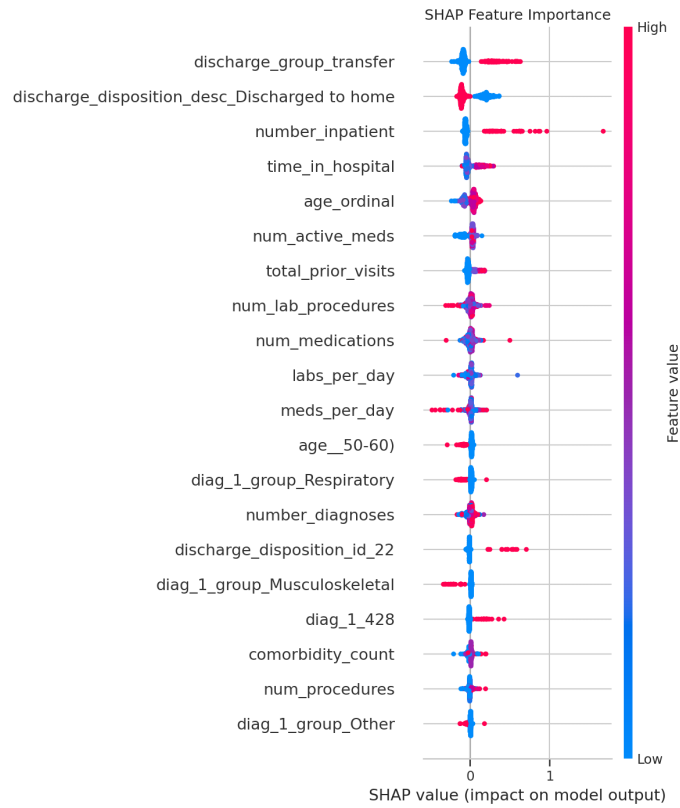


Fig. 4: Global SHAP summary highlighting the most impactful clinical features on readmission likelihood.

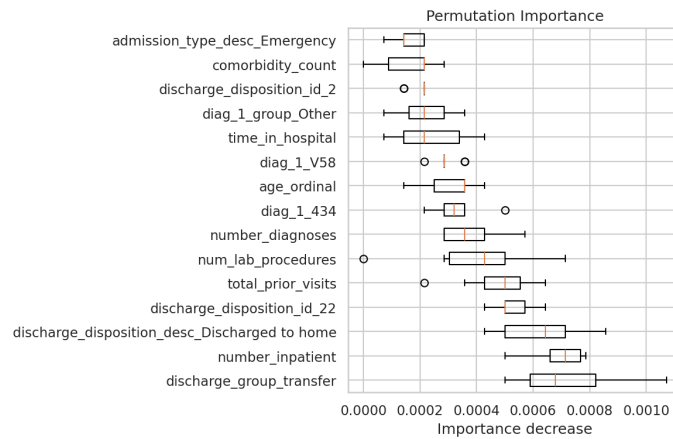


Fig. 5: Permutation importance ranking demonstrating the reliance on historical utilization and medication complexity.

programmatic approach ensures that discharge resources are allocated strategically, deliberately trading inexpensive outpatient care to mitigate severe HRRP financial penalties.

To complement the supervised predictive models, we applied unsupervised learning techniques to discover intrinsic patient phenotypes and isolate high-risk subpopulations within the high-dimensional feature space (over 2,400 one-hot encoded features). Principal Component Analysis (PCA) revealed significant sparsity, requiring 1,598 components to explain 80% of the variance (Figure 9). Using a PCA-50 representation to reduce dimensionality, we applied K-Means and Gaussian Mixture Models (GMM), both confirming an optimal cluster count of $k = 3$ based on silhouette scores and Bayesian Information Criterion (BIC) evaluation (Figure 10).

Profiling these clusters uncovered distinct clinical subgroups (Figure 11 and Table III). While global association rule mining across the entire dataset failed to isolate rules directly predicting early readmission at a minimum support of 0.05, stratifying the analysis by cluster proved highly effective. Specifically, one of the discovered subgroups (Cluster 2, $n = 3,346$) contained 153 unique association rules directly linking specific patient conditions and medication interactions to the 30-day readmission

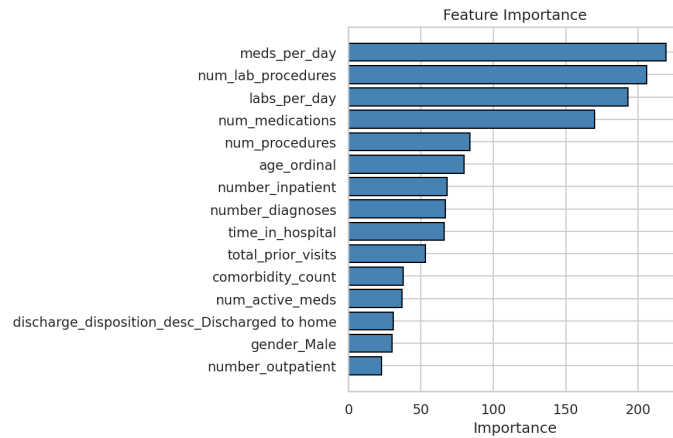


Fig. 6: Feature importance output confirming key drivers of readmission risk.

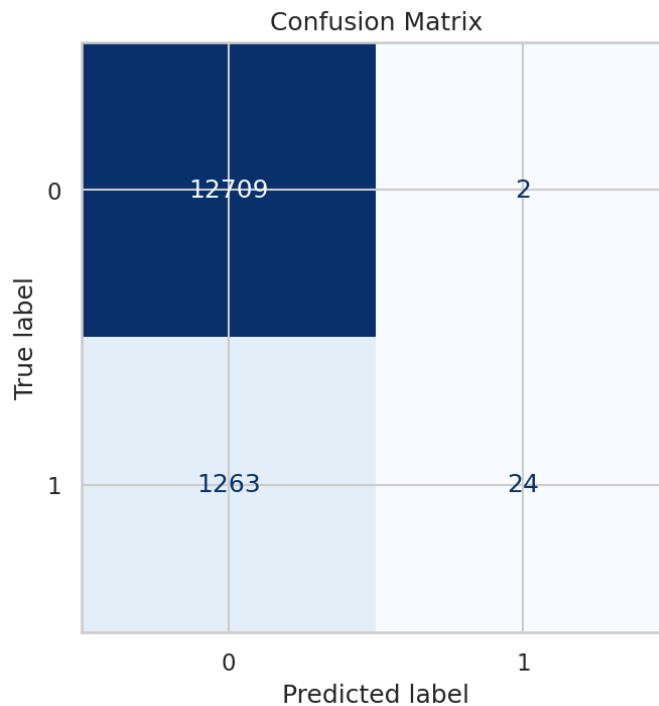


Fig. 7: Confusion matrix evaluated on the test set using the default threshold, visualizing the extreme drop in minority class recall.

consequent. This reinforces that readmission drivers are not strictly uniform across the hospital population but vary drastically between underlying patient phenotypes, allowing for more targeted, cluster-specific intervention criteria compared to generic global rules.

TABLE III: Mean Clinical Profiles Across Discovered Clusters

Cluster	Time in Hospital	Lab Procedures	Medications	Diagnoses
0	4.29	47.7	14.6	7.4
1	4.18	36.7	16.8	7.0
2	4.82	42.0	15.9	6.5

V. CONCLUSIONS AND TAKEAWAYS

In summary, this project shifts the focus of predicting 30-day diabetic readmissions from chasing raw accuracy to emphasizing Cost-Sensitive Knowledge Discovery. By processing the Diabetes 130-US Hospitals dataset safely and effectively treating class

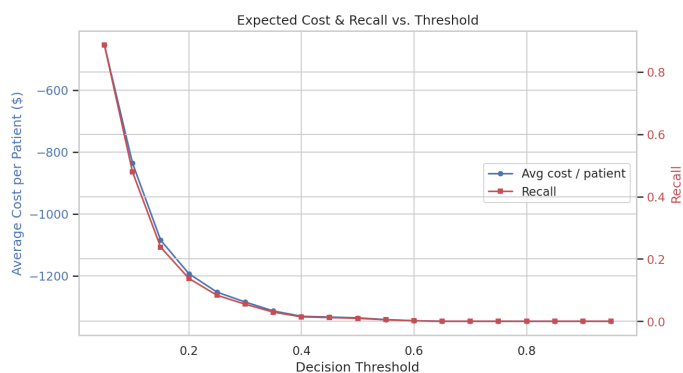


Fig. 8: Cost-threshold sweep demonstrating the optimal probability threshold to minimize financial penalties.

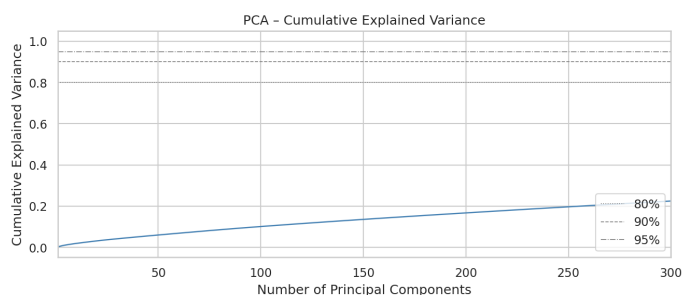


Fig. 9: PCA cumulative explained variance, illustrating the high dimensionality and sparsity of the clinical features.

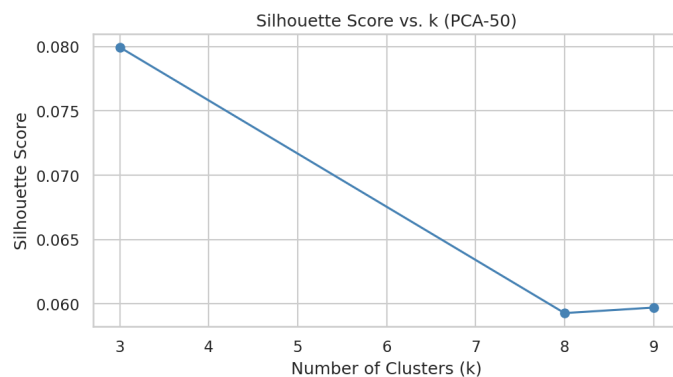


Fig. 10: Silhouette scores for K-Means clustering on the PCA-50 projection, identifying an optimal $k = 3$ patient clusters.

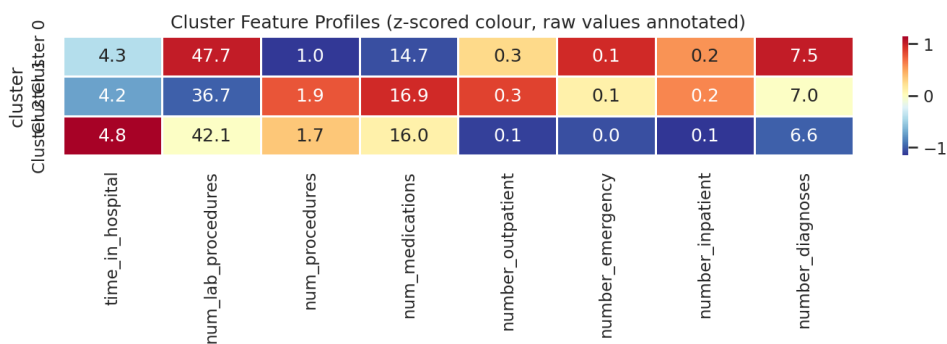


Fig. 11: Feature heatmap highlighting distinct clinical profiles across the three discovered patient clusters.

imbalance via SMOTE inside nested validation loops, we demonstrated how transparent, interpretable processes can approach the utility of ensemble techniques. It should be noted that the dataset was sourced from the United States, hence caution may

be prudent for direct generalization to healthcare systems with different structures. However, the core methodological insights regarding cost-sensitive evaluation and the importance of interpretable models are broadly applicable across healthcare contexts.

Based on the exploration, modeling, and clustering findings, we propose the following actionable recommendations for hospital administration and clinical teams:

- **Adopt Cost-Sensitive Default Thresholds:** A threshold heavily optimized for the HRRP penalty structure (where missing a readmission costs significantly more than preventative intervention) is necessary for any predictive model to be financially viable, and is valuable for imbalanced medical datasets in general which tend to have the positive diagnosis as the minority class.
- **Avoid Reliance on Global Heuristics:** Because the vast majority of hospital patients are low-acuity, global rules and dataset-wide patterns inherently optimize for identifying stable, non-readmitted patients (as demonstrated by our top dataset-wide association rules). Hospitals must avoid depending on broad, universal indicators to capture readmission risk, as they fail to isolate the elusive high-risk minority class.
- **Implement Targeted, Phenotype-Specific Interventions:** Recognize that readmission drivers are not uniform across the patient population. Interventions should be uniquely tailored to specific patient phenotypes. Our clustering identified a high-risk subgroup (Cluster 2) representing only $\sim 5\%$ of patients, yet yielding 153 highly specific readmission rules. Concentrating discharge resources on this small phenotype is significantly more efficient than population-wide screening.
- **Prioritize Transparent Rule Sets Over Black-Box Scores:** Logic, not probabilities, may be more valuable in decision-making contexts. Deploying findings as interpretable IF-THEN rules (like those found via Association Rule Mining in the high-risk cluster) ensures clinical buy-in at the point of discharge.
- **Focus on High-Impact Clinical Drivers:** Build standard clinical EHR alerts around the strongest predictors of readmission found in the exploration and modeling phases. These primarily include prior hospital utilization (`number_inpatient`), the absolute volume of concurrent prescriptions (`num_medications`), and complex comorbidity indicators (e.g., compounding glycemic control abnormalities).

Ultimately, optimizing decision thresholds explicitly against the profound cost disparities of Missed Readmissions (FN) versus Unnecessary Preventative Care (FP) not only mitigates HRRP penalties but directly delivers comprehensible, actionable guidelines for hospital decision-makers.

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